A

Synopsis

on

Smart Expense Tracker

Submitted in partial fulfillment of the requirements

for the iOS Project

in

Computer Science and Engineering

by

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Semester – III

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(Session: 2026–2027)

1. ABSTRACT

Smart Expense Tracker is a SwiftUI-based iOS application developed to help users record, organize and monitor their daily expenses in a simple and structured manner. The application combines expense recording, category-wise organization, monthly budgeting, spending analytics, search, editing and multilingual support in a single mobile application. Persistent storage is handled using SwiftData, while Swift Charts is used to visualize spending patterns.

The application allows users to add expenses with amount, category, note and date; edit or delete existing records; search transactions; set a monthly budget; and monitor remaining budget and spending progress. It supports English, Hindi, Marathi and Urdu and provides a guided onboarding flow. The main aim is to give users a convenient way to understand where their money is being spent and encourage better monthly financial planning.

Keywords: iOS, SwiftUI, SwiftData, Swift Charts, Expense Tracking, Budget Management, Spending Analysis, Mobile Application.

2. INTRODUCTION

Managing small daily expenses can become difficult when spending is recorded inconsistently or not tracked at all. For students, expenses commonly include food, travel, shopping, education and other personal requirements. Without a systematic record, it can be difficult to understand where money is being spent or whether spending is within a planned monthly limit.

Traditional methods such as handwritten records, notes or spreadsheets require manual effort and do not always provide automatic totals, category-wise analysis or budget feedback. A mobile application can simplify this process by keeping expense information organized and accessible from a smartphone.

Smart Expense Tracker provides a centralized platform for recording and organizing expenses. The dashboard summarizes total spending and recent transactions. Expenses can be categorized into Food, Travel, Shopping, Education and Others. The monthly budget module displays spending progress and remaining amount, while the Spending Chart module provides category-wise visual analysis for This Month, Last Month and All Time.

3. PROBLEM FORMULATION

The project addresses the need for a simple iOS application that can record expenses, organize them into categories, monitor budgets, provide visual spending analysis and allow users to manage financial records efficiently. The key problems identified are:

- Difficulty in maintaining a proper record of daily expenses.
- Lack of an organized method to categorize spending.
- Difficulty in monitoring monthly budgets and remaining amounts.
- Lack of visual understanding of spending patterns.
- Difficulty in quickly finding a particular expense.
- Manual records can be time-consuming to maintain.
- Need for a simple interface with multilingual support.

4. OBJECTIVES

- Develop an easy-to-use iOS application for recording and managing daily expenses.
- Provide categorized expense management for Food, Travel, Shopping, Education and Others.
- Provide monthly budget management with clear budget status indicators.
- Allow users to add, edit, search and delete expense records.
- Visualize category-wise spending using Swift Charts.
- Provide multilingual support for English, Hindi, Marathi and Urdu.
- Create a modern, consistent and user-friendly interface.

5. METHODOLOGY

The application follows a SwiftUI-based component structure. The application-level model container provides shared SwiftData access, while dedicated views handle specific user tasks. Stored expense data acts as the source for dashboard totals, budget calculations and chart data so that the displayed information remains synchronized.

Step	Process
1	Launch application and complete onboarding if required.
2	Select language and enter the dashboard.
3	Add an expense with amount, category, note and date.
4	Store the expense using SwiftData.
5	Update totals, budget progress and category analytics.
6	Review spending through Chart and Budget screens.
7	Edit, search or delete records when required.

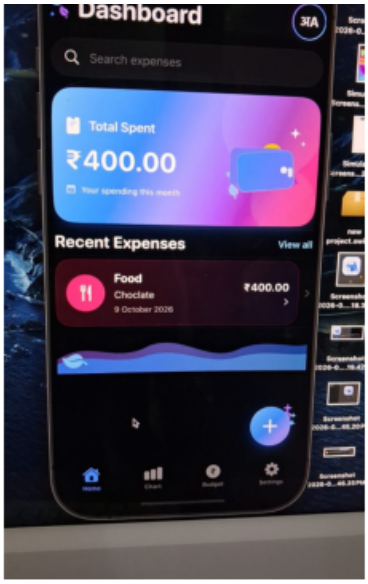
The application is developed using the following technology stack:

Layer	Technology	Purpose
Language	Swift	Application logic and functionality
UI	SwiftUI	Screens, navigation and interface
Persistence	SwiftData	Local storage of expenses and budgets
Charts	Swift Charts	Spending visualization
Development	Xcode	Build, test and debug the application
Platform	iOS	Target mobile platform

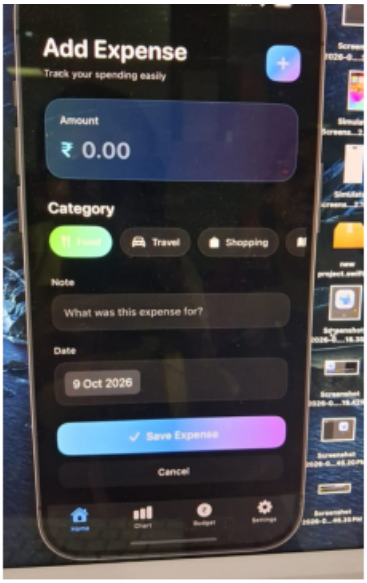
Data requirements: Expense amount, expense category, expense note, expense date, monthly budget amount, selected language and onboarding state.

6. RESULTS

The implemented application provides a consistent black interface with vibrant accent elements, wallet/expense-themed visual identity, clear cards and bottom navigation for Home, Chart, Budget and Settings. The following original pages show the working core modules and additional functional screens.



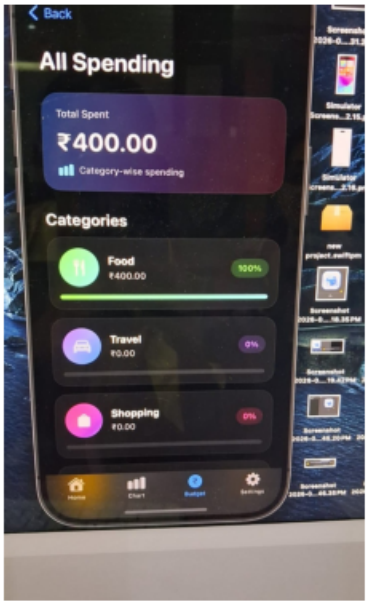
Dashboard



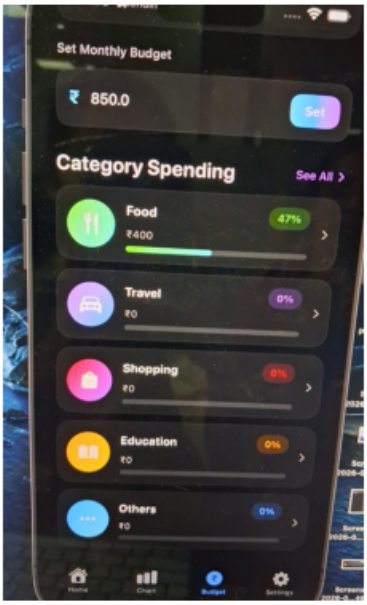
Add Expense



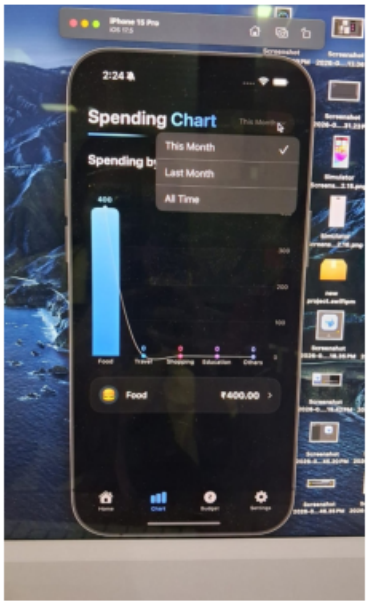
Budget



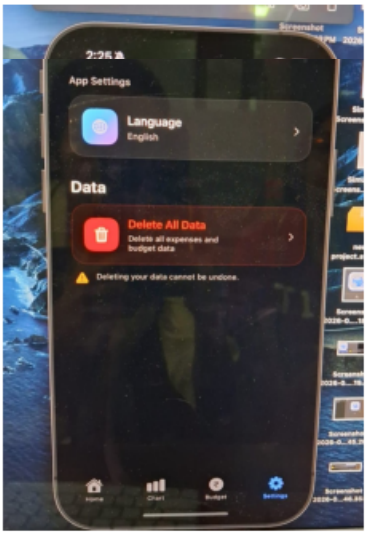
All Spending
All Spending



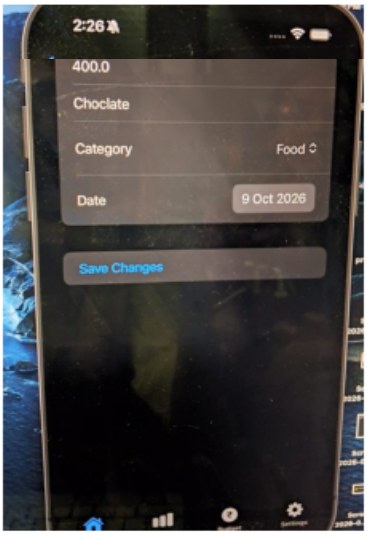
Category Spending
Category Spending



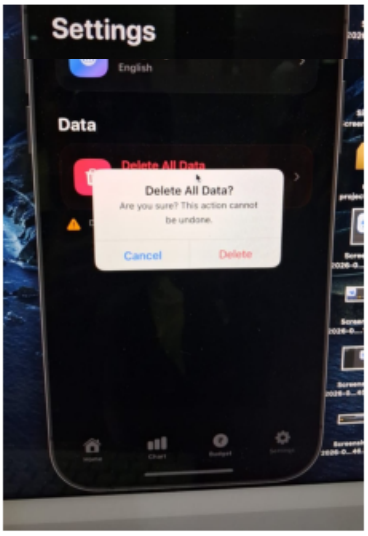
Spending Chart
Spending Chart



Settings: language and data controls.



Edit Expense: update amount, note, category and date.



Delete All Data: confirmation before permanent deletion.

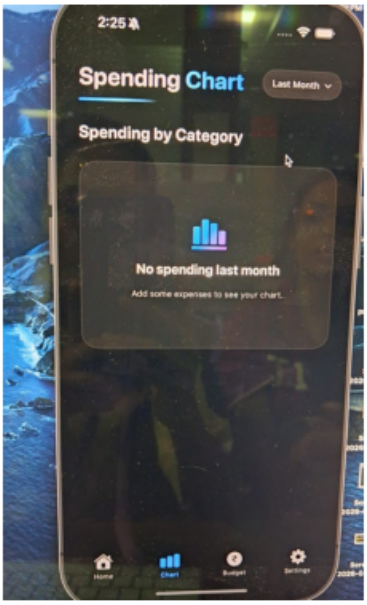
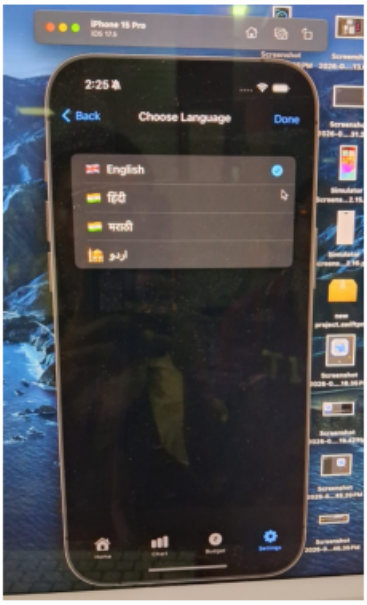
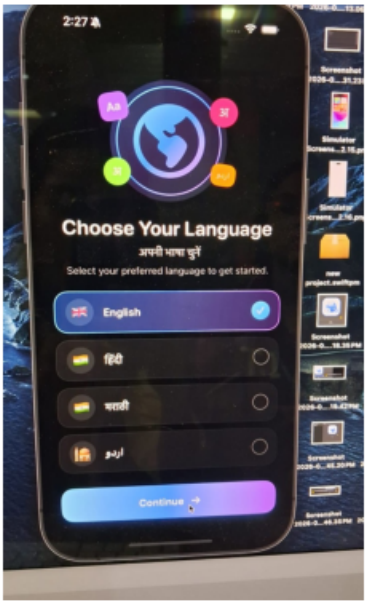


Chart period selection / empty-state handling.

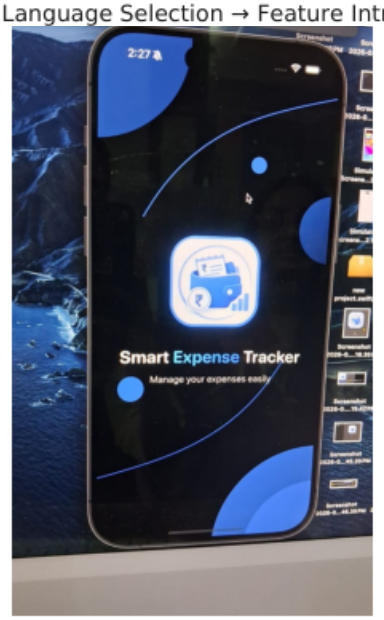


Settings language selection.

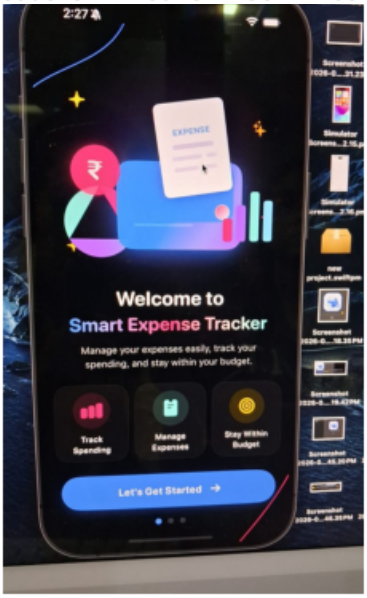


Onboarding language selection.

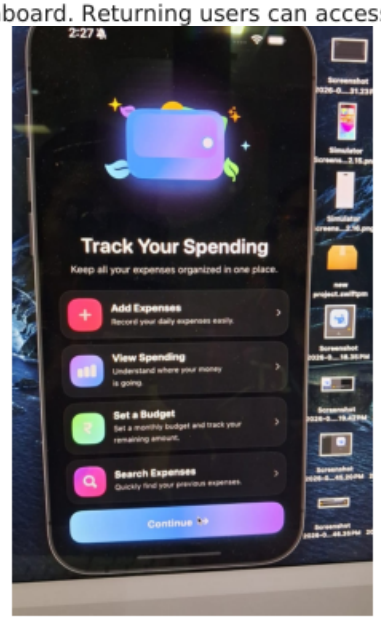
The first-launch experience guides the user through a short sequence: Splash Screen → Welcome → Language Selection → Feature Introduction → You're All Set → Dashboard. Returning users can access



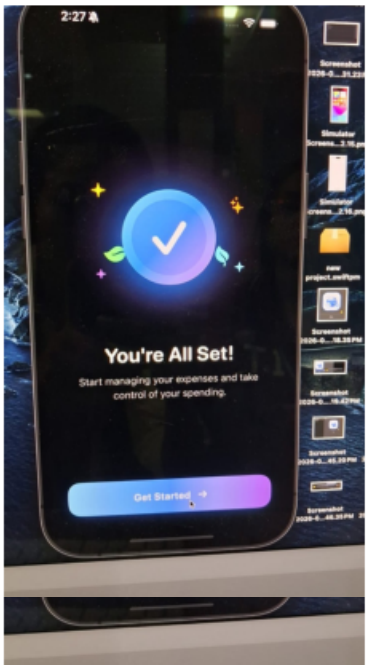
Splash Screen: application identity and launch screen.

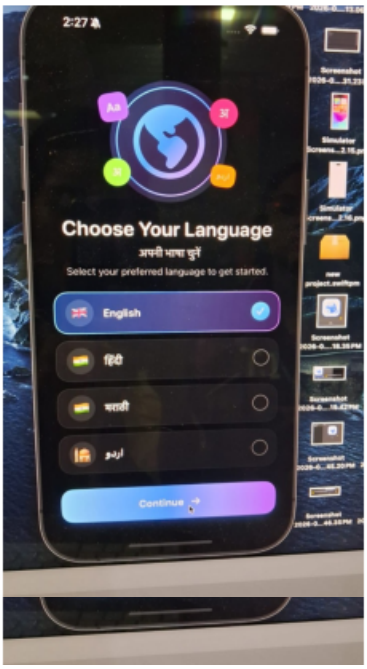


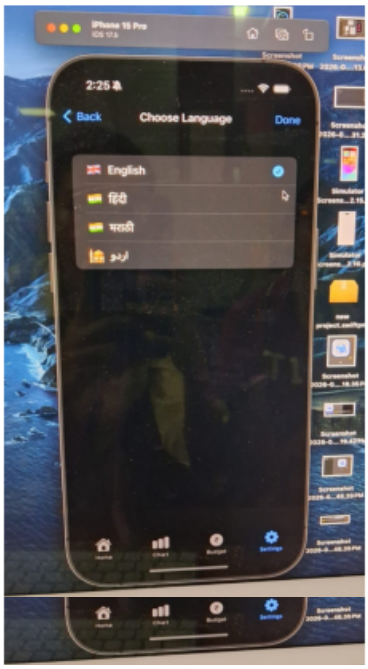
Welcome: purpose, core benefits and entry to setup.



Feature Introduction: Add Expenses, View Spending, Set a Budget and Search Expenses.







7. CONCLUSION

Smart Expense Tracker successfully combines expense recording, budgeting and spending visualization into a single iOS application. Its modular interface and persistent local data make it practical for everyday expense management while also demonstrating important concepts of modern iOS development.

The application provides a clear workflow from onboarding and language selection to adding, editing, searching and deleting expenses, monitoring a monthly budget and reviewing spending through charts. The completed implementation provides a strong foundation that can be extended with cloud services, advanced analytics and secure user accounts in future versions.